2026 03 05
발표 자료

광운대학교 로봇학과
FAIR Lab

김한서

KWANGWOON
UNIVERSITY

이번 주 진행사항

- Towards Accurate Time Series Forecasting via Implicit Decoding
    - 논문 리뷰

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Towards Accurate Time Series Forecasting via Implicit Decoding

Xinyu Li[1], **Yuchen Luo**[1], **Hao Wang**[2], **Haoxuan Li**[3,6],
Liuhua Peng[1], **Feng Liu**[1], **Yandong Guo**[4], **Kun Zhang**[5,6], **Mingming Gong**[1,6*]
[1]The University of Melbourne [2]Zhejiang University [3]Peking University [4]AI[2] Robotics
[5]Carnegie Mellon University [6]Mohamed bin Zayed University of Artificial Intelligence
xl5@student.unimelb.edu.au,mingming.gong@unimelb.edu.au

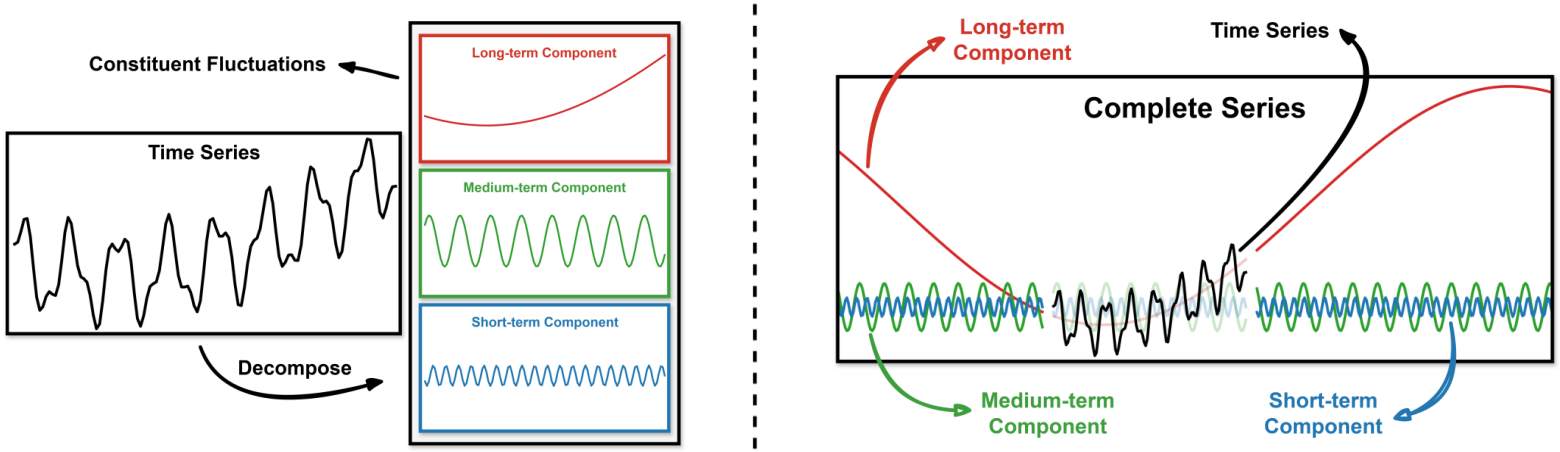
- arXiv 등록일: 2025-09-19
- 인용 수: 9회(Google Scholar, 2026-03-03)
- Published at NeurIPS 2025
- 링크: https://openreview.net/pdf?id=gqoeQPhQcE

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Background



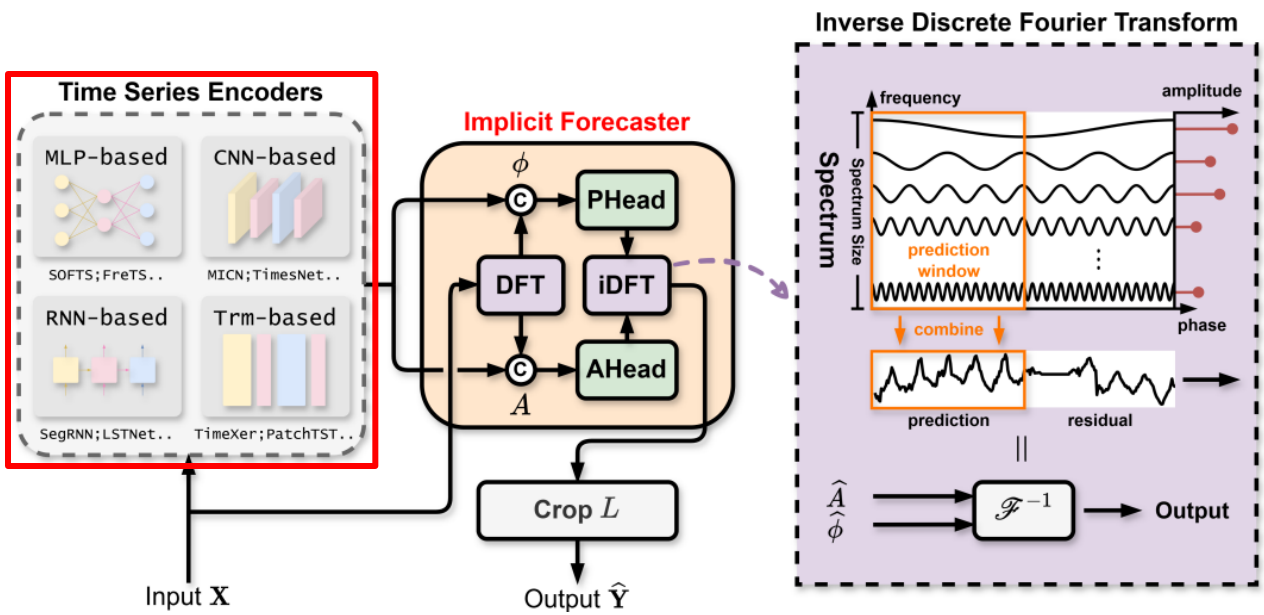
- 기존 시계열 예측 모델들의 개별 시점에 대한 독립적인 예측은 전체적인 흐름이나 상관관계를 충분히 반영하지 못함
- Discrete Fourier Transform을 통해 시계열 데이터를 전체적인 추세 정보인 Long-term, 중간 정도의 주기를 가지는 Medium-term, 계절성 정보인 Short-term Component로 분리하여 처리함

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

모델 구조



- Time Series Encoders
  - Encoder는 기존 모델을 Backbone으로 사용하며, Channel Independence 방식 사용
  - 시계열 데이터를 입력하여 고유 정보와 시간적 특징을 포착할 수 있는 Hidden Representation으로 변환

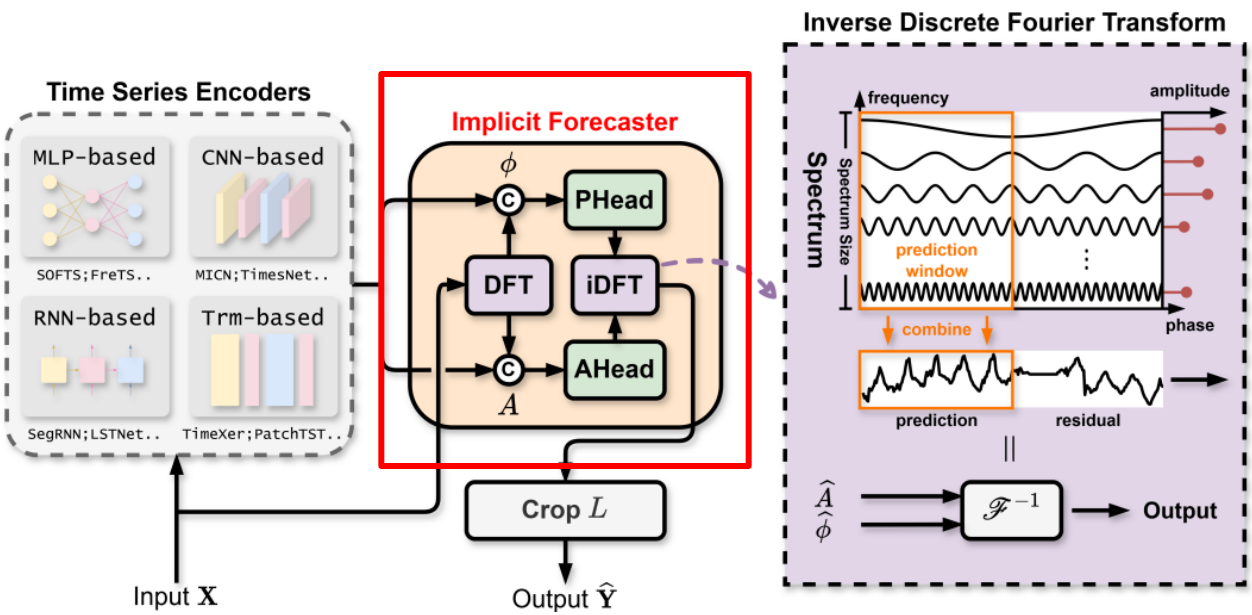
$X_{enc} = \mathrm{TimeSeriesEncoder}(X)$, $\mathrm{TimeSeriesEncoder}: \mathbb{R}^{T\times N} \mapsto \mathbb{R}^{N\times D}$

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

모델 구조



- Implicit Forecaster (DFT, Phead, Ahead)
  - Ahead는 미래 Amplitude를 예측, Phead는 값의 불연속성을 해결하기 위해 2개의 MLP로 sin 성분($\hat{\alpha}$)과 cos 성분($\hat{\beta}$)을 예측한 뒤, 미래 Phase를 예측

$$\mathrm{AHead}(\mathrm{X_{enc,n}}, \mathrm{A_n}) = \left|\mathrm{MLP_{amp}}\left(\mathrm{Concat}(\mathrm{X_{enc,n}}, \mathrm{A_n})\right)\right|,\ \widehat{\alpha_n} = \mathrm{Tanh}\left(\mathrm{MLP_{sin}}\left(\mathrm{Concat}(\mathrm{X_{enc,n}}, \phi_n)\right)\right),\ \widehat{\beta_n} = \mathrm{Tanh}\left(\mathrm{MLP_{cos}}\left(\mathrm{Concat}(\mathrm{X_{enc,n}}, \phi_n)\right)\right),\ \widehat{\phi_n} = atan2(\widehat{\alpha_n}, \widehat{\beta_n})$$

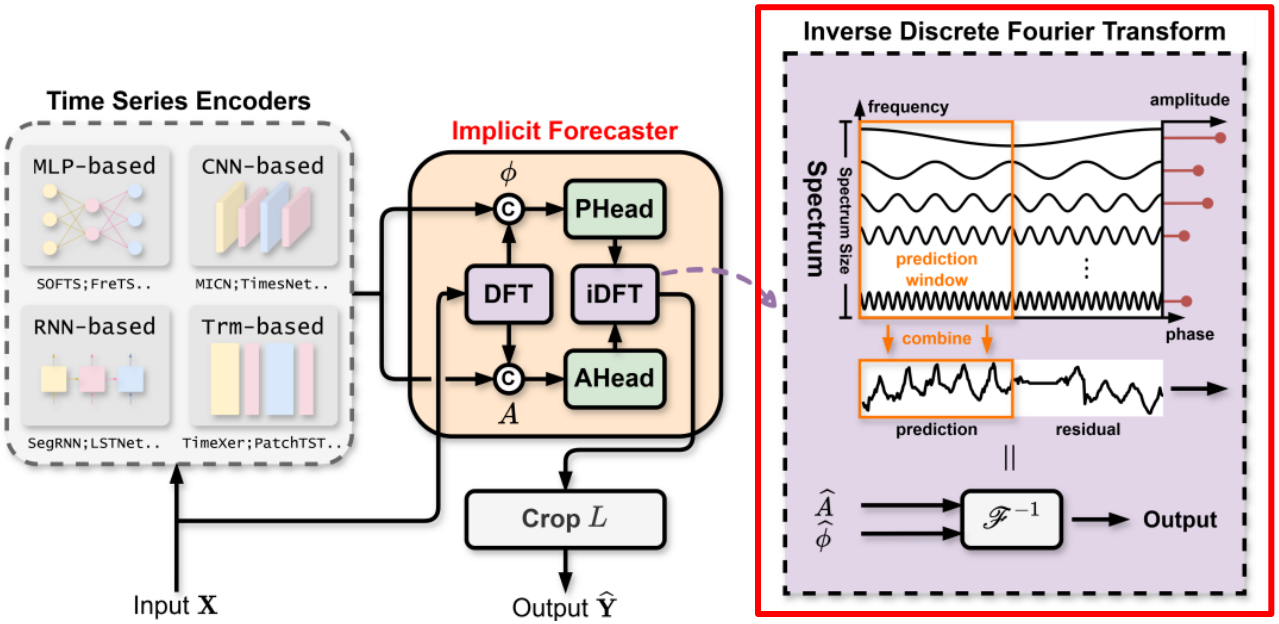
  - Frequency Pool P의 크기를 예측 길이 L의 2배로 설정, 켤레 대칭 원리를 통해 P의 절반만 예측

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

모델 구조



- Implicit Forecaster (iDFT)
  - 각 Head에서 예측한 Amplitude와 Phase를 복소수 형태로 결합 후, 켤레 대칭으로 전체 스펙트럼 완성
  - iDFT를 적용해 시간 도메인으로 복원하고 예측 길이만큼 Crop하여 최종 예측값 생성 $\hat{Y} = Crop_{:L}\left(\mathcal{F}^{-1}\left(\hat{A} \cdot e^{j\hat{\phi}}\right)^{\top}\right)$

$\hat{A} \cdot e^{j\hat{\phi}}$: 예측된 진폭과 위상을 오일러 공식을 통해 복소수 형태로 결합한 것

j: 허수 단위

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

주요 모델 성능 비교

Models	SOFTS (2024)		**w/ IF (Ours)**		TimeXer (2024)		**w/ IF (Ours)**		iTransformer (2024)		**w/ IF (Ours)**		SegRNN (2023)		**w/ IF (Ours)**		PatchTST (2023)		**w/ IF (Ours)**		Crossformer (2023)		**w/ IF (Ours)**		TimesNet (2023)		**w/ IF (Ours)**	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.533	0.519	**0.506**	**0.516**	0.527	**0.525**	**0.510**	0.528	**0.535**	**0.525**	**0.535**	0.529	0.524	**0.523**	**0.523**	0.526	0.533	0.522	**0.498**	**0.511**	0.607	0.570	**0.587**	**0.564**	0.596	0.568	**0.535**	**0.546**
ETTh2	0.202	0.315	**0.193**	**0.305**	0.204	0.317	**0.194**	**0.306**	0.202	0.315	**0.198**	**0.309**	0.193	**0.307**	**0.189**	0.308	0.192	0.308	**0.187**	**0.301**	0.296	0.395	**0.238**	**0.360**	0.212	0.327	**0.204**	**0.315**
ETTm1	0.486	**0.474**	**0.473**	0.475	0.490	0.488	**0.477**	**0.480**	0.487	0.481	**0.478**	**0.478**	**0.470**	0.474	**0.470**	**0.473**	0.477	0.476	**0.471**	**0.470**	0.538	0.516	**0.532**	**0.511**	0.548	0.512	**0.508**	**0.497**
ETTm2	0.151	0.269	**0.147**	**0.266**	**0.145**	0.262	**0.145**	**0.261**	0.154	0.272	**0.150**	**0.268**	**0.142**	**0.259**	0.144	0.263	0.147	0.266	**0.144**	**0.262**	0.203	0.319	**0.183**	**0.304**	0.155	0.271	**0.148**	**0.264**
ECL	0.170	**0.258**	**0.165**	0.261	0.166	0.265	**0.161**	**0.257**	0.172	**0.261**	**0.167**	0.264	**0.181**	**0.275**	0.186	0.277	0.199	0.289	**0.184**	**0.274**	0.212	0.308	**0.193**	**0.294**	0.190	**0.289**	**0.188**	0.291
Traffic	0.433	**0.286**	**0.429**	0.287	**0.480**	**0.297**	0.629	0.343	**0.434**	**0.292**	0.497	0.333	0.639	0.326	**0.573**	**0.315**	0.489	0.317	**0.470**	**0.308**	0.575	**0.326**	**0.546**	0.336	0.614	**0.336**	**0.593**	0.359
Weather	0.236	**0.261**	**0.229**	0.270	**0.221**	**0.254**	**0.221**	0.260	0.237	**0.261**	**0.225**	**0.261**	0.230	0.281	**0.227**	**0.277**	0.237	**0.262**	**0.229**	**0.262**	**0.242**	**0.305**	0.253	0.313	0.241	0.270	**0.231**	**0.269**
Exchange	0.424	0.454	**0.402**	**0.438**	0.397	**0.433**	**0.395**	0.434	0.427	0.458	**0.412**	**0.449**	0.440	0.446	**0.294**	**0.393**	0.441	0.454	**0.422**	**0.446**	**0.798**	**0.654**	1.338	0.825	0.425	0.456	**0.420**	**0.454**
ILI	2.893	1.122	**2.827**	**1.120**	3.134	1.174	**2.938**	**1.147**	2.991	1.164	**2.871**	**1.119**	4.631	1.432	**3.641**	**1.246**	3.365	1.155	**3.185**	**1.127**	**4.246**	**1.440**	5.128	1.592	**3.264**	**1.162**	3.734	1.177
PEMS03	**0.107**	**0.214**	0.109	0.216	0.122	0.235	**0.102**	**0.209**	0.414	0.392	**0.209**	**0.293**	0.196	0.298	**0.159**	**0.270**	0.252	0.341	**0.159**	**0.263**	0.167	0.279	**0.146**	**0.259**	0.175	0.260	**0.106**	**0.208**
PEMS04	0.112	0.223	**0.107**	**0.216**	0.113	0.231	**0.098**	**0.207**	0.187	0.282	**0.124**	**0.234**	0.233	0.328	**0.191**	**0.296**	0.334	0.393	**0.186**	**0.290**	0.232	0.331	**0.132**	**0.249**	0.222	0.325	**0.117**	**0.232**
PEMS07	0.096	**0.195**	**0.092**	**0.195**	0.100	0.205	**0.078**	**0.178**	0.884	0.629	**0.149**	**0.249**	0.211	0.302	**0.174**	**0.272**	0.275	0.351	**0.173**	**0.267**	0.196	0.290	**0.155**	**0.259**	0.167	0.280	**0.109**	**0.215**
PEMS08	0.118	0.218	**0.107**	**0.212**	0.144	0.260	**0.108**	**0.217**	0.860	0.626	**0.179**	**0.284**	0.232	0.320	**0.176**	**0.282**	0.301	0.371	**0.172**	**0.276**	0.220	0.317	**0.142**	**0.257**	0.210	0.315	**0.128**	**0.242**
Solar	0.222	**0.253**	**0.207**	**0.253**	0.239	0.290	**0.231**	**0.272**	0.241	0.278	**0.228**	**0.271**	0.244	0.298	**0.238**	**0.286**	0.240	0.288	**0.229**	**0.272**	0.299	0.344	**0.222**	**0.283**	0.260	0.273	**0.212**	**0.270**
Promotion	—	—	**3.71%**	**0.60%**	—	—	**4.97%**	**3.54%**	—	—	**18.5%**	**11.0%**	—	—	**10.4%**	**5.12%**	—	—	**14.5%**	**8.82%**	—	—	**6.62%**	**4.07%**	—	—	**14.3%**	**7.22%**

- 기존 모델들에 Implicit Forecaster를 적용했을 때, MSE와 MAE 모두 성능 향상이 된 것을 확인하였음

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Ablation Study

Transformer	Decoder							
Replace	**IF**		Linear		MLP		Transformer	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh2	**0.188**	**0.301**	0.201	0.313	0.199	0.311	0.202	0.316
Exchange	**0.286**	**0.384**	0.411	0.442	0.428	0.448	0.412	0.454
ILI	**2.387**	**1.018**	2.689	1.107	2.486	1.040	3.504	1.282
PEMS08	**0.103**	**0.209**	0.184	0.272	0.198	0.278	0.167	0.272
Performance	**100%**	**100%**	77.0%	88.0%	77.3%	88.9%	73.1%	84.0%

IF	Skip-connection					
Replace	**w/ Skip.**		w/o Skip.		only Skip.	
Metric	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	**0.503**	**0.516**	0.537	0.521	0.617	0.577
ECL	**0.157**	0.256	0.160	**0.254**	0.210	0.300
Weather	**0.221**	0.259	0.227	**0.257**	0.250	0.291
Solar	**0.189**	**0.251**	0.190	0.251	0.284	0.345
Performance	**100%**	99.9%	**97.2%**	**100%**	**77.8%**	**84.0%**

- 기존 모델들의 Decoder보다 Implicit Forecaster를 사용했을 때가 가장 좋은 성능을 보임
- Skip-connection을 통해 원본 데이터의 정보를 활용할 때가 가장 좋은 성능을 보임

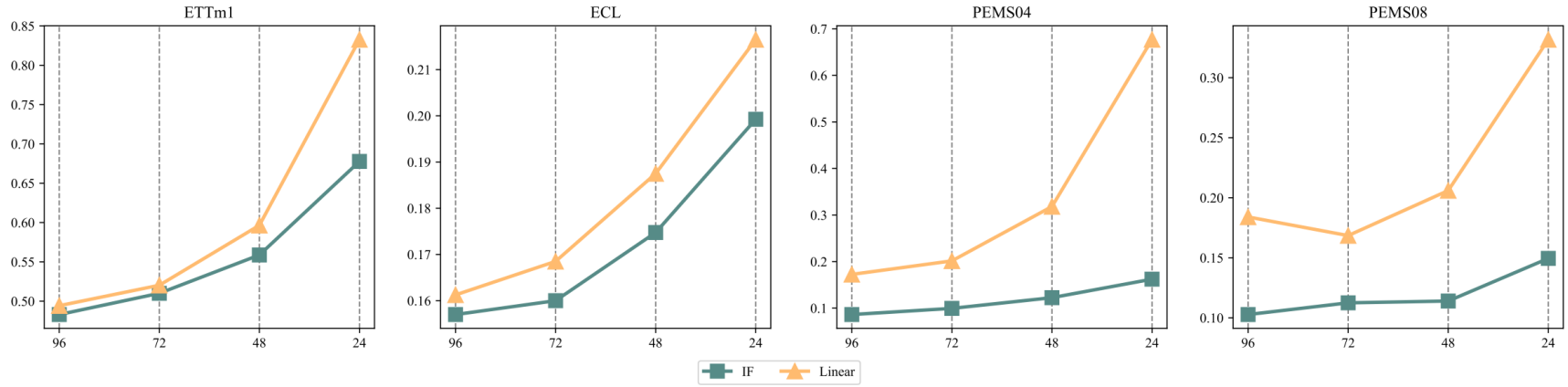
부록

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Varying lookback length



- 96부터 24까지의 입력 길이에서 Implicit Forecaster가 기존 Linear 방식보다 더 뛰어난 성능을 보임

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Dataset details

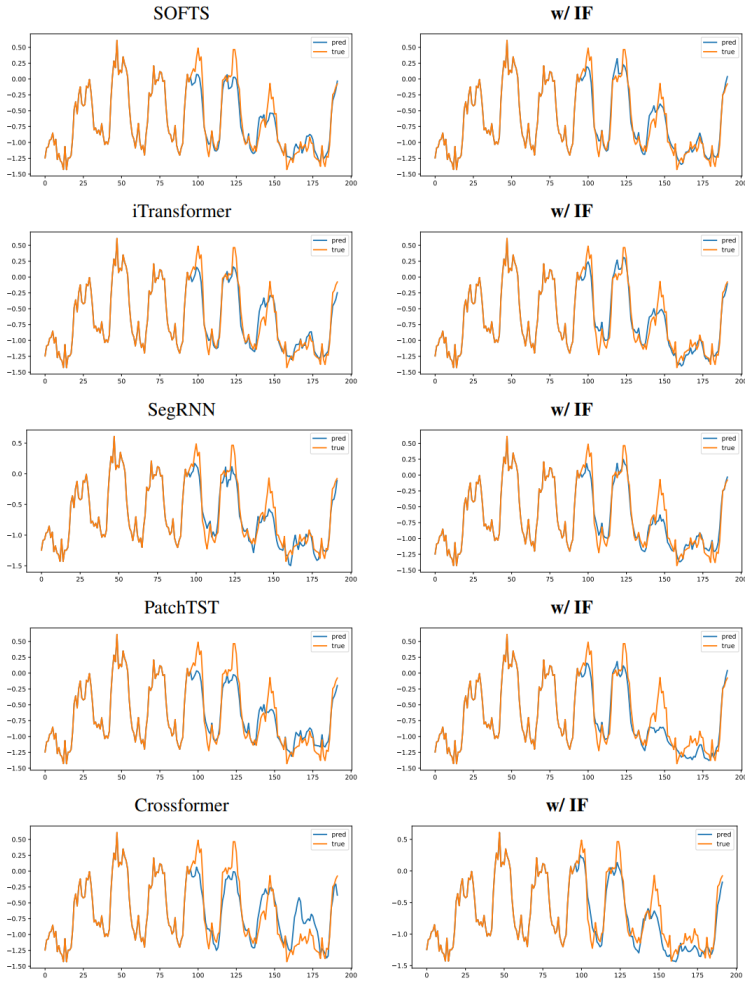
Dataset	Dim	Input Length	Prediction Length	Dataset Size	Frequency	Domain
ETTh1, ETTh2	7	96	$\{96, 192, 336, 720\}$	$(12194, 2613, 2613)$	Hourly	Temperature
ETTm1, ETTm2	7	96	$\{96, 192, 336, 720\}$	$(48776, 10452, 10452)$	15min	Temperature
Electricity	321	96	$\{96, 192, 336, 720\}$	$(18412, 3945, 3947)$	Hourly	Electricity
Traffic	862	96	$\{96, 192, 336, 720\}$	$(12280, 2631, 2633)$	Hourly	Transportation
Weather	21	96	$\{96, 192, 336, 720\}$	$(36887, 7904, 7905)$	10min	Weather
Exchange Rate	8	96	$\{96, 192, 336, 720\}$	$(5311, 1138, 1139)$	Daily	Finance
Illness	7	36	$\{24, 36, 48, 60\}$	$(676, 144, 146)$	Weekly	Health
PEMS03	358	96	$\{12, 24, 48, 96\}$	$(18345, 3931, 3932)$	5min	Transportation
PEMS04	307	96	$\{12, 24, 48, 96\}$	$(11894, 2548, 2550)$	5min	Transportation
PEMS07	883	96	$\{12, 24, 48, 96\}$	$(19756, 4233, 4235)$	5min	Transportation
PEMS08	170	96	$\{12, 24, 48, 96\}$	$(12499, 2678, 2679)$	5min	Transportation
Solar-Energy	137	96	$\{96, 192, 336, 720\}$	$(36792, 7884, 7884)$	10min	Energy

Implicit Forecaster

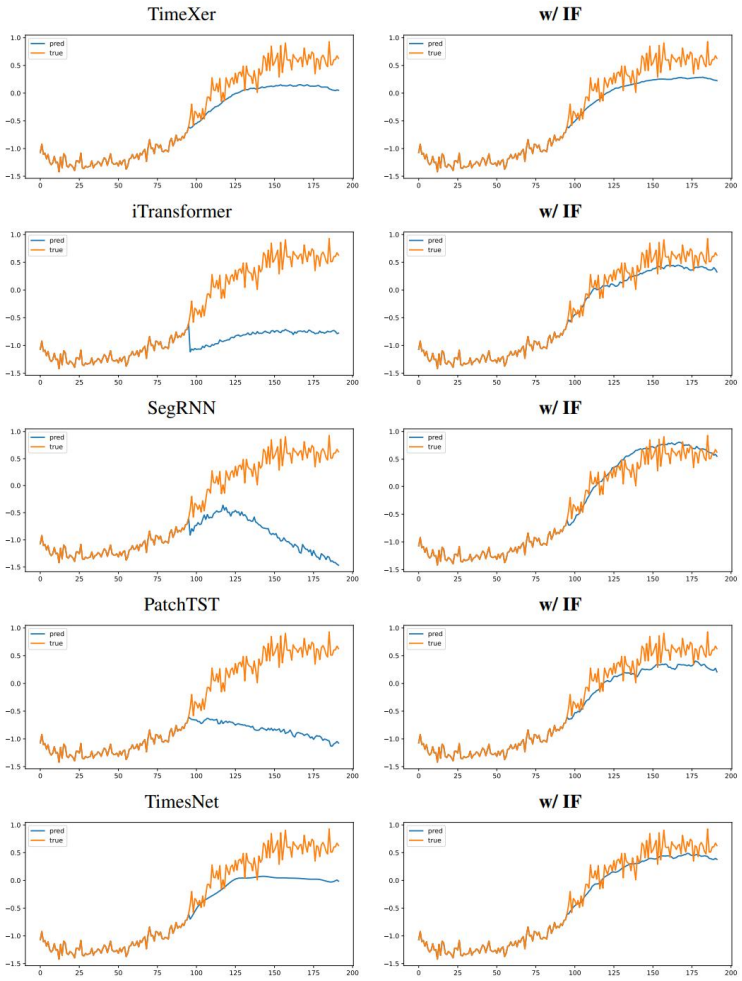
FAIR Lab

KWANGWOON UNIVERSITY

Model Prediction Visualization



Input-96-output-96, ECL dataset



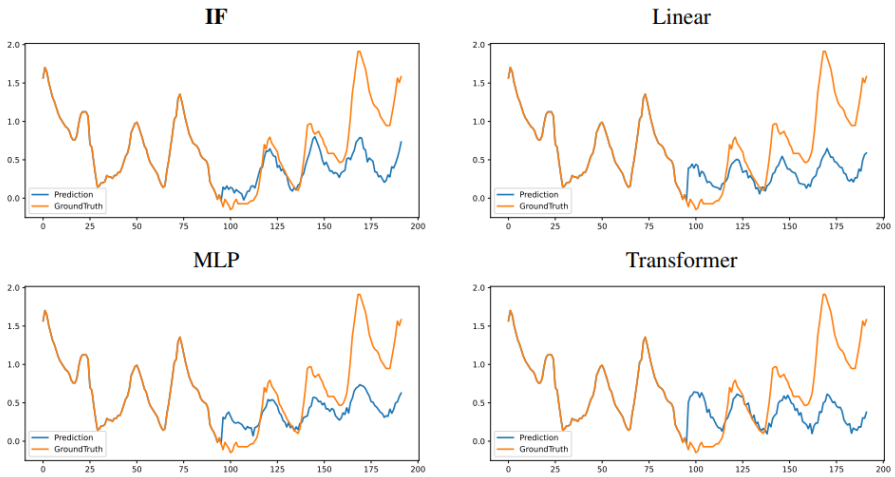
Input-96-output-96, PEMS03 dataset

Implicit Forecaster

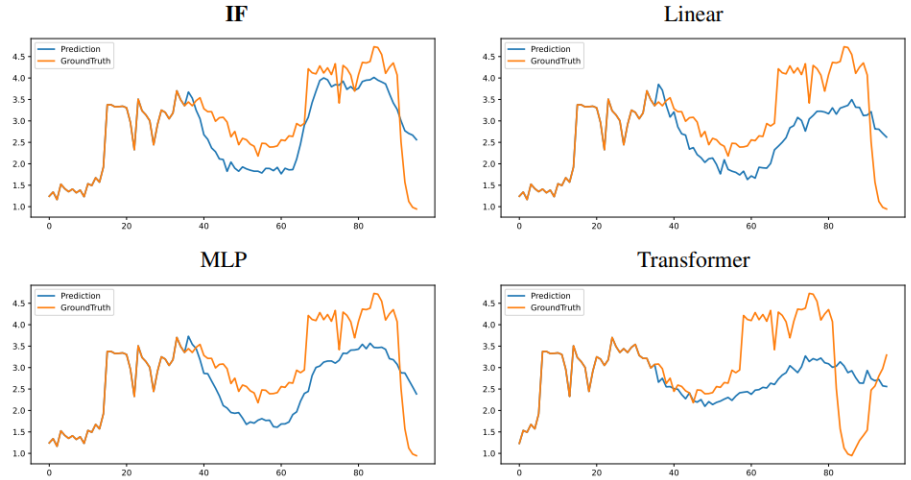
FAIR Lab

KWANGWOON UNIVERSITY

Decoder Prediction Visualization



Input-96-output-96, ETTh2 dataset



Input-36-output-60, ILI dataset

Implicit Forecaster

FAIR Lab

KWANGWOON UNIVERSITY

Full Results

Models		SOFTS (2024)		w/ IF (Ours)		TimeXer (2024)		w/ IF (Ours)		iTransformer (2024)		w/ IF (Ours)		SegRNN (2023)		w/ IF (Ours)		PatchTST (2023)		w/ IF (Ours)		Crossformer (2023)		w/ IF (Ours)		TimesNet (2023)		w/ IF (Ours)	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.442	0.461	**0.426**	**0.458**	0.423	**0.457**	**0.420**	0.461	**0.447**	**0.469**	0.456	0.474	**0.432**	**0.463**	0.436	0.467	0.425	0.457	**0.411**	**0.453**	**0.437**	**0.476**	0.437	**0.474**	0.484	0.501	**0.433**	**0.475**
	192	0.493	0.495	**0.469**	**0.489**	0.487	0.503	**0.461**	**0.494**	0.495	**0.503**	**0.490**	**0.503**	0.480	0.497	**0.467**	**0.490**	**0.477**	**0.494**	0.485	0.500	**0.487**	**0.505**	0.581	0.563	0.542	0.536	**0.498**	**0.521**
	336	0.529	**0.519**	**0.510**	0.523	0.529	**0.528**	**0.524**	0.541	0.537	**0.526**	**0.533**	0.532	0.524	0.526	**0.502**	**0.517**	0.552	0.530	**0.497**	**0.511**	**0.640**	0.602	0.644	**0.590**	0.606	0.576	**0.548**	**0.553**
	720	0.667	0.602	**0.618**	**0.595**	0.668	**0.612**	**0.637**	0.618	0.662	**0.601**	**0.660**	0.608	**0.659**	**0.608**	0.689	0.631	0.678	0.607	**0.601**	**0.582**	0.862	0.696	**0.684**	**0.630**	0.752	0.657	**0.662**	**0.637**
	Avg	0.533	0.519	**0.506**	**0.516**	0.527	**0.525**	**0.510**	0.528	**0.535**	**0.525**	**0.535**	0.529	0.524	**0.523**	**0.523**	0.526	0.533	0.522	**0.498**	**0.511**	0.607	0.570	**0.587**	**0.564**	0.596	0.568	**0.535**	**0.546**
ETTh2	96	0.159	0.279	**0.154**	**0.273**	0.153	0.273	**0.151**	**0.270**	0.160	0.280	**0.159**	**0.277**	0.149	**0.269**	**0.148**	0.270	0.152	0.272	**0.147**	**0.266**	0.179	0.310	**0.171**	**0.302**	0.168	0.290	**0.161**	**0.279**
	192	0.188	0.304	**0.182**	**0.295**	0.184	0.301	**0.177**	**0.292**	0.187	0.303	**0.186**	**0.300**	0.180	0.296	**0.175**	**0.295**	0.183	0.302	**0.179**	**0.295**	0.261	0.381	**0.223**	**0.353**	0.209	0.323	**0.199**	**0.309**
	336	0.208	0.320	**0.195**	**0.309**	0.215	0.327	**0.209**	**0.322**	0.208	0.322	**0.202**	**0.315**	0.200	0.315	**0.192**	**0.312**	0.198	0.316	**0.194**	**0.309**	0.256	0.384	**0.251**	**0.371**	0.216	0.332	**0.210**	**0.322**
	720	0.254	0.356	**0.240**	**0.342**	0.263	0.367	**0.239**	**0.341**	0.253	0.356	**0.244**	**0.346**	**0.242**	**0.350**	0.243	0.356	0.234	0.343	**0.227**	**0.333**	0.487	0.505	**0.306**	**0.413**	0.257	0.361	**0.246**	**0.349**
	Avg	0.202	0.315	**0.193**	**0.305**	0.204	0.317	**0.194**	**0.306**	0.202	0.315	**0.198**	**0.309**	0.193	**0.307**	**0.189**	0.308	0.192	0.308	**0.187**	**0.301**	0.296	0.395	**0.238**	**0.360**	0.212	0.327	**0.204**	**0.315**
ETTm1	96	**0.398**	**0.416**	0.406	0.425	**0.417**	**0.432**	0.435	0.438	**0.392**	0.426	0.396	**0.425**	0.400	0.424	**0.389**	**0.416**	**0.390**	0.418	0.403	**0.417**	0.497	0.488	**0.487**	**0.467**	0.487	0.469	**0.463**	**0.460**
	192	**0.438**	**0.448**	0.441	0.454	0.458	0.469	**0.446**	**0.458**	0.447	**0.458**	**0.443**	**0.458**	0.438	0.455	**0.435**	**0.450**	**0.445**	0.455	**0.445**	**0.452**	0.515	0.497	**0.479**	**0.480**	0.514	0.490	**0.478**	**0.476**
	336	0.520	0.492	**0.488**	**0.485**	0.509	0.503	**0.495**	**0.495**	0.523	0.498	**0.499**	**0.493**	**0.492**	0.489	0.495	**0.488**	0.504	0.490	**0.490**	**0.483**	0.541	0.524	**0.531**	**0.517**	0.573	0.534	**0.517**	**0.505**
	720	0.589	0.540	**0.555**	**0.537**	0.577	0.550	**0.534**	**0.529**	0.585	0.540	**0.572**	**0.538**	**0.548**	**0.529**	0.562	0.536	0.571	0.540	**0.544**	**0.527**	**0.600**	**0.554**	0.633	0.580	0.620	0.557	**0.572**	**0.548**
	Avg	0.486	**0.474**	**0.473**	0.475	0.490	0.488	**0.477**	**0.480**	0.487	0.481	**0.478**	**0.478**	**0.470**	0.474	**0.470**	**0.473**	0.477	0.476	**0.471**	**0.470**	0.538	0.516	**0.532**	**0.511**	0.548	0.512	**0.508**	**0.497**
ETTm2	96	0.109	0.227	**0.101**	**0.220**	**0.098**	**0.217**	0.099	**0.217**	0.107	0.227	**0.105**	**0.225**	**0.097**	**0.216**	0.100	0.217	0.102	0.222	**0.100**	**0.219**	0.112	0.240	**0.105**	**0.227**	0.106	0.225	**0.101**	**0.218**
	192	0.130	0.252	**0.128**	**0.249**	0.125	0.247	**0.124**	**0.244**	0.132	0.255	**0.130**	**0.253**	**0.123**	**0.244**	0.125	0.246	0.128	0.251	**0.127**	**0.250**	0.147	0.267	**0.137**	**0.262**	0.136	0.254	**0.130**	**0.248**
	336	0.162	0.280	**0.158**	**0.279**	**0.155**	**0.273**	0.156	**0.273**	0.168	0.286	**0.161**	**0.280**	**0.151**	**0.270**	0.154	0.274	0.158	0.279	**0.152**	**0.272**	0.238	0.344	**0.209**	**0.338**	0.165	0.281	**0.158**	**0.276**
	720	0.204	0.315	**0.202**	**0.314**	0.201	0.311	**0.200**	**0.309**	0.209	0.319	**0.205**	**0.315**	**0.195**	**0.306**	0.198	0.313	0.199	0.310	**0.195**	**0.308**	0.315	0.425	**0.282**	**0.391**	0.215	0.322	**0.203**	**0.313**
	Avg	0.151	0.269	**0.147**	**0.266**	**0.145**	0.262	**0.145**	**0.261**	0.154	0.272	**0.150**	**0.268**	**0.142**	**0.259**	0.144	0.263	0.147	0.266	**0.144**	**0.262**	0.203	0.319	**0.183**	**0.304**	0.155	0.271	**0.148**	**0.264**
ECL	96	0.146	**0.233**	**0.141**	**0.233**	0.142	0.242	**0.137**	**0.231**	0.150	**0.239**	**0.145**	**0.239**	**0.157**	**0.249**	0.165	0.254	0.179	0.271	**0.163**	**0.250**	0.164	0.266	**0.161**	**0.263**	0.167	0.268	**0.160**	**0.265**
	192	**0.157**	**0.245**	**0.157**	0.250	0.155	0.252	**0.151**	**0.247**	0.160	**0.248**	**0.158**	0.253	**0.169**	**0.260**	0.174	0.263	0.184	0.275	**0.170**	**0.259**	0.186	0.284	**0.176**	**0.277**	0.182	0.282	**0.178**	**0.281**
	336	0.170	**0.259**	**0.166**	0.263	0.171	0.271	**0.165**	**0.261**	0.172	**0.263**	**0.169**	0.268	**0.181**	**0.277**	0.185	0.278	0.197	0.289	**0.182**	**0.274**	0.213	0.308	**0.200**	**0.303**	**0.192**	**0.293**	0.194	0.298
	720	0.208	**0.294**	**0.197**	0.296	0.198	0.295	**0.190**	**0.290**	0.208	**0.295**	**0.196**	**0.295**	**0.216**	**0.312**	0.219	**0.312**	0.234	0.321	**0.221**	**0.314**	0.285	0.376	**0.236**	**0.335**	**0.218**	**0.315**	**0.218**	0.320
	Avg	0.170	**0.258**	**0.165**	0.261	0.166	0.265	**0.161**	**0.257**	0.172	**0.261**	**0.167**	0.264	**0.181**	**0.275**	0.186	0.277	0.199	0.289	**0.184**	**0.274**	0.212	0.308	**0.193**	**0.294**	0.190	**0.289**	**0.188**	0.291
Traffic	96	0.393	**0.267**	**0.390**	**0.267**	**0.438**	**0.278**	0.690	0.341	**0.401**	**0.277**	0.432	0.302	0.609	0.313	**0.530**	**0.298**	0.463	0.304	**0.439**	**0.293**	0.544	**0.311**	**0.512**	0.319	0.579	**0.322**	**0.573**	0.348
	192	0.426	0.281	**0.420**	**0.280**	**0.471**	**0.296**	0.664	0.347	**0.427**	**0.288**	0.458	0.311	0.626	0.319	**0.559**	**0.306**	0.478	0.311	**0.460**	**0.302**	0.565	**0.320**	**0.525**	0.329	0.608	**0.332**	**0.584**	0.363
	336	0.444	**0.289**	**0.440**	0.291	**0.487**	**0.298**	0.621	0.347	**0.441**	**0.294**	0.482	0.324	0.644	0.326	**0.571**	**0.316**	0.493	0.317	**0.475**	**0.308**	0.579	**0.328**	**0.548**	0.337	0.622	**0.338**	**0.592**	0.350
	720	0.468	**0.307**	**0.467**	0.309	**0.523**	**0.316**	0.541	0.338	**0.467**	**0.310**	0.617	0.396	0.675	0.345	**0.633**	**0.339**	0.520	0.335	**0.507**	**0.331**	0.610	**0.346**	**0.598**	0.358	0.648	**0.351**	**0.623**	0.376
	Avg	0.433	**0.286**	**0.429**	0.287	**0.480**	**0.297**	0.629	0.343	**0.434**	**0.292**	0.497	0.333	0.639	0.326	**0.573**	**0.315**	0.489	0.317	**0.470**	**0.308**	0.575	**0.326**	**0.546**	0.336	0.614	**0.336**	**0.593**	0.359
Weather	96	0.164	**0.205**	**0.161**	0.214	0.147	**0.194**	**0.145**	0.196	0.164	0.202	**0.154**	**0.201**	**0.156**	**0.217**	0.160	0.218	0.164	**0.205**	**0.158**	0.206	**0.163**	**0.230**	0.164	0.233	0.165	0.213	**0.156**	**0.210**
	192	**0.204**	**0.240**	0.207	0.257	0.191	**0.234**	**0.189**	0.238	0.207	**0.240**	**0.198**	0.242	**0.198**	0.258	0.199	**0.255**	0.207	**0.242**	**0.196**	**0.242**	**0.222**	0.295	0.224	**0.292**	0.207	**0.249**	**0.203**	0.250
	336	0.257	0.278	**0.241**	**0.277**	**0.244**	**0.273**	0.247	0.282	0.259	0.280	**0.244**	**0.277**	0.251	0.300	**0.245**	**0.294**	0.258	0.279	**0.249**	**0.278**	**0.256**	**0.320**	0.263	0.323	0.272	0.293	**0.255**	**0.289**
	720	0.319	**0.322**	**0.309**	0.331	0.304	**0.314**	**0.303**	0.324	0.319	**0.322**	**0.305**	0.323	0.317	0.348	**0.305**	**0.340**	0.320	**0.320**	**0.312**	0.321	**0.327**	**0.377**	0.359	0.404	0.322	0.327	**0.309**	**0.326**
	Avg	0.236	**0.261**	**0.229**	0.270	**0.221**	**0.254**	**0.221**	0.260	0.237	**0.261**	**0.225**	**0.261**	0.230	0.281	**0.227**	**0.277**	0.237	**0.262**	**0.229**	**0.262**	**0.242**	**0.305**	0.253	0.313	0.241	0.270	**0.231**	**0.269**
Exchange	96	0.089	0.214	**0.082**	**0.204**	0.086	0.207	**0.082**	**0.204**	0.103	0.231	**0.102**	**0.229**	**0.082**	**0.200**	0.085	0.208	0.091	0.212	**0.086**	**0.206**	**0.236**	**0.352**	0.262	0.372	0.096	0.225	**0.090**	**0.217**
	192	0.200	0.328	**0.177**	**0.305**	0.187	0.312	**0.177**	**0.306**	0.206	0.334	**0.188**	**0.318**	**0.174**	**0.298**	0.207	0.333	0.184	0.310	**0.183**	**0.309**	**0.427**	**0.484**	0.526	0.539	**0.185**	**0.319**	0.187	0.321
	336	0.389	0.465	**0.349**	**0.437**	**0.371**	**0.449**	0.373	0.452	**0.362**	**0.448**	0.364	0.450	0.360	0.437	**0.306**	**0.424**	0.361	0.445	**0.345**	**0.434**	**1.063**	**0.807**	2.276	1.191	**0.381**	**0.456**	0.382	0.460
	720	1.017	0.809	**0.999**	**0.807**	**0.943**	**0.763**	0.950	0.773	1.038	0.820	**0.994**	**0.798**	1.142	0.848	**0.578**	**0.608**	1.128	0.847	**1.074**	**0.834**	**1.466**	**0.973**	2.287	1.198	1.037	0.825	**1.023**	**0.816**
	Avg	0.424	0.454	**0.402**	**0.438**	0.397	**0.433**	**0.395**	0.434	0.427	0.458	**0.412**	**0.449**	0.440	0.446	**0.294**	**0.393**	0.441	0.454	**0.422**	**0.446**	**0.798**	**0.654**	1.338	0.825	0.425	0.456	**0.420**	**0.454**
ILI	96	2.731	**1.001**	**2.621**	1.035	**2.903**	1.134	2.941	**1.124**	2.820	1.085	**2.543**	**1.014**	3.974	1.272	**2.920**	**1.078**	2.585	1.005	**2.292**	**1.003**	**3.922**	**1.375**	4.598	1.508	**2.715**	**1.054**	2.811	1.057
	192	**2.665**	**1.105**	2.837	1.121	3.052	1.153	**2.677**	**1.119**	3.034	1.173	**2.880**	**1.113**	4.366	1.379	**3.167**	**1.173**	4.506	1.292	**4.131**	**1.252**	**3.971**	**1.383**	5.034	1.573	**3.670**	**1.245**	5.866	1.333
	336	3.485	1.280	**3.212**	**1.212**	3.402	1.235	**2.926**	**1.154**	**3.158**	**1.224**	3.263	1.232	4.425	1.427	**3.932**	**1.339**	**3.578**	1.219	3.736	**1.202**	**3.958**	**1.420**	4.861	1.543	3.633	1.229	**3.537**	**1.228**
	720	2.691	**1.101**	**2.637**	1.111	**3.179**	**1.174**	3.208	1.191	2.953	1.175	**2.797**	**1.116**	5.759	1.651	**4.547**	**1.395**	2.792	1.102	**2.581**	**1.050**	**5.132**	**1.582**	6.021	1.745	3.037	1.120	**2.723**	**1.089**
	Avg	2.893	1.122	**2.827**	**1.120**	3.134	1.174	**2.938**	**1.147**	2.991	1.164	**2.871**	**1.119**	4.631	1.432	**3.641**	**1.246**	3.365	1.155	**3.185**	**1.127**	**4.246**	**1.440**	5.128	1.592	**3.264**	**1.162**	3.734	1.177
PEMS03	12	0.064	0.167	**0.061**	**0.164**	0.064	0.177	**0.058**	**0.165**	0.065	0.172	**0.062**	**0.169**	0.078	0.191	**0.074**	**0.185**	0.092	0.208	**0.070**	**0.178**	0.083	0.194	**0.073**	**0.185**	0.087	0.195	**0.069**	**0.173**
	24	0.081	**0.189**	**0.080**	**0.189**	0.086	0.203	**0.072**	**0.181**	0.093	0.206	**0.085**	**0.197**	0.123	0.243	**0.113**	**0.229**	0.144	0.266	**0.105**	**0.218**	0.117	0.237	**0.104**	**0.224**	0.118	0.222	**0.094**	**0.192**
	48	**0.118**	**0.230**	0.127	0.238	0.130	0.248	**0.110**	**0.220**	**0.160**	**0.273**	0.279	0.360	0.225	0.333	**0.189**	**0.301**	0.275	0.372	**0.177**	**0.285**	0.207	0.321	**0.174**	**0.287**	0.186	0.272	**0.119**	**0.220**
	96	**0.165**	**0.272**	0.167	0.275	0.207	0.311	**0.167**	**0.269**	1.337	0.918	**0.410**	**0.447**	0.356	0.425	**0.261**	**0.364**	0.496	0.518	**0.285**	**0.370**	0.260	0.363	**0.234**	**0.340**	0.309	0.352	**0.142**	**0.248**
	Avg	**0.107**	**0.214**	0.109	0.216	0.122	0.235	**0.102**	**0.209**	0.414	0.392	**0.209**	**0.293**	0.196	0.298	**0.159**	**0.270**	0.252	0.341	**0.159**	**0.263**	0.167	0.279	**0.146**	**0.259**	0.175	0.260	**0.106**	**0.208**
PEMS04	12	**0.080**	**0.185**	0.085	0.193	0.079	0.192	**0.072**	**0.175**	0.094	0.199	**0.084**	**0.189**	0.103	0.215	**0.097**	**0.208**	0.123	0.242	**0.096**	**0.205**	0.095	0.210	**0.088**	**0.200**	0.135	0.251	**0.082**	**0.190**
	24	0.102	0.213	**0.093**	**0.199**	0.095	0.212	**0.085**	**0.193**	0.130	0.236	**0.102**	**0.211**	0.157	0.273	**0.144**	**0.258**	0.196	0.309	**0.139**	**0.252**	0.126	0.248	**0.111**	**0.230**	0.180	0.290	**0.095**	**0.208**
	48	0.125	0.238	**0.116**	**0.227**	0.120	0.241	**0.100**	**0.211**	0.197	0.296	**0.134**	**0.247**	0.265	0.363	**0.225**	**0.330**	0.371	0.430	**0.217**	**0.321**	0.176	0.300	**0.143**	**0.264**	0.240	0.342	**0.125**	**0.243**
	96	0.141	0.254	**0.136**	**0.247**	0.157	0.279	**0.136**	**0.250**	0.328	0.398	**0.178**	**0.288**	0.406	0.462	**0.298**	**0.390**	0.647	0.591	**0.294**	**0.384**	0.529	0.567	**0.184**	**0.301**	0.335	0.418	**0.166**	**0.288**
	Avg	0.112	0.223	**0.107**	**0.216**	0.113	0.231	**0.098**	**0.207**	0.187	0.282	**0.124**	**0.234**	0.233	0.328	**0.191**	**0.296**	0.334	0.393	**0.186**	**0.290**	0.232	0.331	**0.132**	**0.249**	0.222	0.325	**0.117**	**0.232**
PEMS07	12	0.067	**0.163**	**0.062**	**0.163**	0.064	0.169	**0.055**	**0.150**	0.067	0.167	**0.060**	**0.158**	0.079	0.187	**0.073**	**0.176**	0.089	0.209	**0.072**	**0.176**	0.097	0.204	**0.076**	**0.188**	0.105	0.222	**0.068**	**0.169**
	24	0.083	0.185	**0.074**	**0.174**	0.084	0.193	**0.069**	**0.170**	0.098	0.201	**0.081**	**0.185**	0.130	0.244	**0.117**	**0.227**	0.152	0.270	**0.116**	**0.225**	0.141	0.252	**0.113**	**0.227**	0.130	0.248	**0.090**	**0.191**
	48	0.103	**0.204**	**0.102**	0.207	0.108	0.212	**0.080**	**0.181**	1.511	1.012	**0.205**	**0.309**	0.241	0.338	**0.206**	**0.306**	0.303	0.385	**0.201**	**0.297**	0.255	0.331	**0.178**	**0.282**	0.179	0.294	**0.116**	**0.227**
	96	**0.130**	**0.230**	0.131	0.237	0.142	0.247	**0.107**	**0.210**	1.860	1.138	**0.250**	**0.346**	0.393	0.439	**0.300**	**0.380**	0.556	0.539	**0.302**	**0.370**	0.291	0.374	**0.255**	**0.339**	0.253	0.355	**0.161**	**0.273**
	Avg	0.096	**0.195**	**0.092**	**0.195**	0.100	0.205	**0.078**	**0.178**	0.884	0.629	**0.149**	**0.249**	0.211	0.302	**0.174**	**0.272**	0.275	0.351	**0.173**	**0.267**	0.196	0.290	**0.155**	**0.259**	0.167	0.280	**0.109**	**0.215**
PEMS08	12	0.076	0.179	**0.070**	**0.173**	0.093	0.211	**0.074**	**0.184**	0.087	0.193	**0.078**	**0.184**	0.093	0.203	**0.085**	**0.192**	0.102	0.218	**0.082**	**0.188**	0.098	0.205	**0.085**	**0.195**	0.125	0.240	**0.081**	**0.189**
	24	0.107	0.211	**0.098**	**0.207**	0.134	0.256	**0.101**	**0.217**	0.132	0.242	**0.110**	**0.220**	0.145	0.258	**0.127**	**0.240**	0.168	0.279	**0.122**	**0.231**	0.145	0.260	**0.116**	**0.231**	0.161	0.277	**0.098**	**0.210**
	48	0.153	0.248	**0.123**	**0.230**	0.140	0.256	**0.100**	**0.208**	1.649	1.082	**0.228**	**0.335**	0.253	0.351	**0.202**	**0.311**	0.321	0.406	**0.199**	**0.305**	0.197	0.311	**0.153**	**0.271**	0.219	0.326	**0.134**	**0.251**
	96	**0.136**	**0.236**	0.137	0.237	0.210	0.318	**0.156**	**0.261**	1.571	0.986	**0.302**	**0.397**	0.437	0.467	**0.290**	**0.385**	0.614	0.581	**0.287**	**0.379**	0.439	0.493	**0.216**	**0.331**	0.335	0.416	**0.199**	**0.318**
	Avg	0.118	0.218	**0.107**	**0.212**	0.144	0.260	**0.108**	**0.217**	0.860	0.626	**0.179**	**0.284**	0.232	0.320	**0.176**	**0.282**	0.301	0.371	**0.172**	**0.276**	0.220	0.317	**0.142**	**0.257**	0.210	0.315	**0.128**	**0.242**
Solar	96	0.192	0.233	**0.185**	**0.231**	0.207	0.268	**0.202**	**0.250**	0.210	0.253	**0.202**	**0.243**	0.226	0.290	**0.218**	**0.273**	0.211	0.271	**0.210**	**0.256**	**0.193**	**0.257**	**0.193**	0.259	0.211	**0.236**	**0.198**	0.253
	192	0.223	**0.253**	**0.215**	0.255	0.241	0.293	**0.240**	**0.276**	0.240	0.277	**0.225**	**0.268**	0.244	0.304	**0.240**	**0.288**	0.247	0.294	**0.235**	**0.272**	0.231	**0.281**	**0.224**	0.287	0.256	0.281	**0.221**	**0.274**
	336	0.238	0.263	**0.218**	**0.262**	0.257	0.303	**0.243**	**0.282**	0.258	0.291	**0.246**	**0.287**	0.259	0.304	**0.248**	**0.291**	0.257	0.296	**0.237**	**0.281**	0.249	0.298	**0.236**	**0.293**	0.288	0.282	**0.222**	**0.280**
	720	0.234	0.264	**0.210**	**0.263**	0.251	0.297	**0.237**	**0.281**	0.255	0.291	**0.238**	**0.285**	0.249	0.295	**0.245**	**0.291**	0.246	0.293	**0.233**	**0.278**	0.522	0.539	**0.235**	**0.294**	0.285	0.295	**0.218**	**0.271**
	Avg	0.222	**0.253**	**0.207**	**0.253**	0.239	0.290	**0.231**	**0.272**	0.241	0.278	**0.228**	**0.271**	0.244	0.298	**0.238**	**0.286**	0.240	0.288	**0.229**	**0.272**	0.299	0.344	**0.222**	**0.283**	0.260	0.273	**0.212**	**0.270**